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The Hamilton Anxiety Scale: reliability, validity and sensitivity to change in anxiety and depressive disorders

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Summary

The Hamilton Anxiety Scale (HAM-A) was tested for reliability and validity in two different samples, one sample ($n = 97$) defined by anxiety disorders, the other sample ($n = 101$) defined by depressive disorders. The reliability and the concurrent validity of the HAM-A and its subscales proved to be sufficient. Internal validity tested by latent structure analysis was insufficient. The major problems with the HAM-A are that (1) anxiolytic and antidepressant effects cannot be clearly distinguished; (2) the subscale of somatic anxiety is strongly related to somatic side effects. The applicability of the HAM-A in anxiolytic treatment studies is therefore limited. More specific anxiety scales are needed.

Key words: Anxiety; Depression; Rating scale

Introduction

The Hamilton Anxiety Scale (HAM-A, Hamilton, 1969) was presented as a rating scale for the severity of anxiety neurosis. Hamilton (1969) considered his anxiety scale only as a proposal open to change during the process of validation. Today, the original version of the HAM-A is mainly used as the indicator for anxiolytic efficacy in nearly all evaluation studies with potential anxiolytic drugs. No rival observer scale for the severity of anxiety has so far received comparable acceptance. In

contrast to this importance of the HAM-A, the number of validation studies is small (Kellner et al., 1968; Guy, 1976; Gjerris et al., 1982; Snaith et al., 1982). These validation studies give only limited support to the validity of the HAM-A.

The HAM-A was designed for patients with a diagnosis of anxiety neurosis and the validation was mainly performed for this group of patients. However, diagnostic habits have changed and the concept of anxiety neurosis has been substituted by a new concept of anxiety disorders comprising panic disorder, phobic disorders and generalized anxiety disorder. The degree of overlap between the old and new concepts is limited (Maier et al., 1986) and therefore the HAM-A is often applied beyond its intended scope. Consequently, new

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validation studies for the HAM-A applied in panic and phobic disorders and generalized anxiety disorder are necessary.

The HAM-A is used for the evaluation of anxiolytic drugs, which are prescribed for the treatment of not only anxiety neurosis or anxiety disorders but also anxiety syndromes arising during other disorders (mainly depressive disorders). The HAM-A is commonly used for this purpose because no other reliable scale is available. However, the HAM-A was not designed and has not been validated for this field of application. The reliability and validity of the HAM-A in depressive disorders will therefore be one of the topics of this paper.

The available validity studies focus mainly on the reliability and selected aspects of cross-sectional validity, especially concurrent validity (i.e., the extent of agreement with global ratings or other assessments of the severity of anxiety). According to the theory of Nunnally (1972), multiple aspects of validity are relevant:

- (1) internal validity (i.e., the extent to which the items of a scale refer to the same stable construct operationalized in the scale);
- (2) content validity (i.e., the extent to which all items of the scale refer to the construct to be operationalized);
- (3) external validity (i.e., the extent to which the scale predicts other measures which are external to the scale but which are connected with the content of the scale by accepted theories, e.g., discrimination between different diagnostic categories or course-related validity).

Investigating the validity of a scale requires the specification of the scope of application. Course-related validity must be evidenced because the main field of application is anxiolytic treatment trials: the HAM-A must be able to detect true intraindividual change in the severity of anxiety states. No generally accepted biological indicators of response to anxiolytics are available. Therefore, the only criterion for identifying true individual change in the severity of anxiety is the expert judgment of the treating physician. This type of validity is another major topic of this paper.

The HAM-A was used in the version of Bech et al. (1986), who reformulated the items and each category of each item in a more precise way.

Patients and methods

Patients and assessments

Sample 1 consisted of 97 consecutive in- and outpatients with panic disorder (DSM-III-R) first contacting a psychiatric or medical center for treatment of their anxiety syndrome. Three clinical psychiatrists performed the SCID (Spitzer and Williams, 1984) and subsequently asked open questions in order to rate assessment scales for the severity of anxiety and depression including the HAM-A; the raters carried out a common training of 15 sessions before starting the study. Diagnoses and global ratings of anxiety by the Covi scale (Lipman, 1982) were performed independently of the assessment of the HAM-A by another psychiatrist conducting a structured interview and subsequently asking open questions. Forty-five patients had a major depressive episode (DSM-III-R) in addition to the panic disorder or agoraphobia. Twenty-nine of the patients in this sample received either imipramine or alprazolam during a 3-week period. The treating psychiatrist documented the status of the patient each week on the basis of an extensive interview and a diary for a daily self-rating of symptoms by the patient. The treating psychiatrist also performed a weekly global rating of the severity of anxiety (Covi scale); independent of the global rating, the HAM-A was rated weekly.

Sample 2 consisted of 101 consecutive inpatients with a major depressive episode; 28 of these patients in addition had panic disorder or agoraphobia according to DSM-III-R (American Psychiatric Association, 1985); patients were interviewed by two raters simultaneously using a comprehensive structured interview including the SCID to which the diagnoses refer; subsequently, open questions were asked in order to rate some assessment scales for the severity of depression and anxiety including the HAM-A, the Raskin Depression Scale and the Covi Anxiety Scale (Lipman, 1981). The raters ($n = 10$) were clinical psychiatrists who had had at least 15 training sessions before starting interviewing and rating; the pairs of rating psychiatrists rotated from patient to patient. In this sample, 45 patients, including 16 with panic attacks, were treated in a standardized manner: they were allocated to fixed-dose im-

ipramine or amitriptyline treatment for 3 weeks (randomized allocation); the treating psychiatrist carefully documented the psychiatric status of the patients daily and rated the change during the 3-week treatment weekly by using the Covi scale and the Raskin scale, which is a scale for the assessment of the severity of depression (Lipman, 1981). Somatic side effects of the treatment were rated with the AMDP system (Guy and Ban, 1980). Only symptoms not present at the beginning and developing during the treatment are counted as side effects. Independently, the HAM-A was rated weekly by another clinician.

The raters in both samples participated in a training program of 15 sessions; after the training, the reliability of the global score of the HAM-A was higher than 0.70 for each rater.

Statistical analysis

Interrater reliability was assessed by intraclass coefficients (Bartko and Carpenter, 1973); this coefficient controls agreement due to chance.

The scores in different groups are compared by nonparametric procedures (*U*-test for 2 groups, Kruskal–Wallis for 3 groups). Correlations between different scores are measured by the Spearman coefficient. Course validity requires the measures of change based on cross-sectional measurements; analysis of covariance was used for deriving change scores (residuals).

Internal validity is assessed by Rasch analysis (Bech, 1981); this latent structure analysis was carried out by separating the sample on the basis of different sets of binary criteria in different ways (e.g., young/old, male/female, associated depressive disorder/no depressive disorder in sample 1); for each pair of subsamples, the consistency of the estimated coefficients in each subsample is tested by a Chi-square statistic (Andersen, 1972). Significant results indicate insufficient homogeneity and lack of transferability of the scale across different subsamples.

Results

Reliability

In sample 2, joint rater interviews were performed; the two ratings for each patient were carried out independently. The reliability of the HAM-A was examined by correlating the parallel

TABLE 1

INTRACLAS COEFFICIENTS (ICC) FOR THE ITEMS AND TOTAL SCORES OF THE HAM-A (JOINT RATER INTERVIEW) (SAMPLE 1).

Item No.	Item content	ICC
1	Anxious mood	0.61
2	Tension	0.37
3	Fears	0.68
4	Insomnia	0.49
5	Intellectual retardation	0.66
6	Depressed mood	0.66
7	General somatic symptoms (muscular)	0.43
8	General somatic symptoms (sensory)	0.52
9	Cardiovascular symptoms	0.77
10	Respiratory symptoms	0.28
11	Gastrointestinal symptoms	0.63
12	Genito-urinary symptoms	0.48
13	Autonomic symptoms	0.23
14	Behavior at interview	0.24
Total scale score (items 1–14)		0.74
Total subscale score (psychic anxiety)		0.73
Total subscale score (somatic anxiety)		0.70

assessments. The interrater reliability was measured by the intraclass coefficient with the following results (Table 1): HAM-A total score: 0.74; HAM-A total score of psychic anxiety: 0.73; HAM-A total score of somatic anxiety: 0.70. Items 10 (respiratory symptoms), 14 (behavior at interview) and 13 (autonomic symptoms) had the lowest reliability (lower than 0.30).

Concurrent validity

The Spearman coefficients in sample 1 between the total score of the Covi Anxiety Scale and the total score of the HAM-A is 0.63 (Table 2), with the total score of somatic anxiety 0.60 and the total score of psychic anxiety 0.69. The Spearman coefficient in sample 2 between the global rating and the total HAM-A score is 0.75, 0.80 for the total score of psychic anxiety, and 0.85 for the total score of somatic anxiety. These coefficients differ significantly from zero ($P < 0.001$).

In both samples, neither the total score of the HAM-A nor the total scores of the subscales of psychic and somatic anxiety deviated from the normal distribution (Kolmogoroff–Smirnov test, $P < 0.05$).

TABLE 2

CONCURRENT VALIDITY OF THE HAM-A

Spearman correlation coefficient between the HAM-A scores and the global assessment of anxiety in sample 1 and sample 2 (HAM-A and global assessments are performed independently).

Item No.	Item content	Spearman correlation	
		Sample 1	Sample 2
1	Anxious mood	0.63	0.67
2	Tension	0.60	0.66
3	Fear	0.49	0.55
4	Insomnia	0.26	0.45
5	Intellectual retardation	0.39	0.39
6	Depressed mood	0.53	0.30
7	General somatic symptoms (muscular)	0.26	0.39
8	General somatic symptoms (sensory)	0.42	0.48
9	Cardiovascular symptoms	0.59	0.63
10	Respiratory symptoms	0.36	0.40
11	Gastrointestinal symptoms	0.34	0.10 *
12	Genito-urinary symptoms	0.39	0.15 *
13	Autonomic symptoms	0.50	0.40
14	Behavior at interview	0.62	0.67
Total scale score		0.63	0.75
Total subscale score (psychic anxiety)		0.69	0.80
Total subscale score (somatic anxiety)		0.60	0.85

* Not significantly different from zero (*t*-test, $P < 0.01$).

Internal validity by latent structure analysis

Latent structure analysis (Rasch analysis) was applied to the HAM-A (total scale, scales for psychic anxiety and somatic anxiety) in both samples (Table 3). In both samples an insufficient Rasch-model fit of the total HAM-A was observed for several procedures for subdividing into pairs of subsamples ($P < 0.01$). Items disturbing the homogeneity and transferability of the original HAM-A scale most frequently are Nos. 3 (fears), 6 (depressed mood), 11 (gastrointestinal symptoms) and 14 (behavior at interview) in the sample of anxious patients and Nos. 3 (fears), 11 (gastrointestinal symptoms) and 13 (autonomic symptoms) in the sample of depressed patients; these items showed an insufficient Rasch fit ($P < 0.05$) for at least two ways of subdividing the sample. How-

ever, the HAM-A subscales for psychic anxiety and for somatic anxiety fit the Rasch model in all ways of subdividing in both samples ($P > 0.01$, Table 3).

Validity by separating diagnostic subtypes

In the sample of anxious patients, the HAM-A total score and the subscores for psychic and somatic anxiety were able to discriminate between patient with panic disorder and co-occurring agoraphobia or major depressive episode on the one hand and with panic disorder but without these additional syndromes on the other hand ($P < 0.05$) (Table 4). In the sample of depressed patients, neither the HAM-A total score nor the subscores for psychic and somatic anxiety were able to discriminate between patients with and without additional DSM-III-R (American Psychiatric Association, 1985) syndromes of anxiety ($P < 0.05$) (Table 4).

Validity by sensitivity to change

The HAM-A scores reflect a significant improvement during the 3-week treatment in both samples (Table 5). The only exception is the subscale for somatic anxiety, which did not improve significantly in the sample of depressed patients (sample 2) (Table 5). In both samples, items 8, 10, 11 and 12 did not improve significantly ($P > 0.05$) during the 3-week treatment.

In both treatment trials, the anxiolytic effect measured by the HAM-A total score and the HAM-A psychic subscale score is highly correlated with the globally assessed anxiolytic effect (Covi scale) (0.59 and 0.73) (Table 6). In sample 2 (depressed patients) both HAM-A scores are equally highly correlated with the antidepressant effect assessed by the Raskin scale (0.58 and 0.73) (Table 6); in the sample of anxious patients, no instruments measuring the antidepressant effect were used.

In the sample of depressed patients, the association between anxiolytic effect (Covi scale) and antidepressant effect (Raskin scale) was 0.46 (correlation between the residuals which are controlled for the initial scores); the analogous correlation coefficient between the HAM-A total score and the Raskin scale score is higher (0.58) (Table 6).

TABLE 3

INTERNAL VALIDITY OF THE HAM-A AND ITS SUBSCALES TESTED BY RASCH ANALYSIS

Empirical *P* values for Rasch fit; *P* < 0.01 indicates insufficient Rasch fit.

Procedure of partition	Sample and scales					
	Sample 1 (defined by anxiety)			Sample 2 (defined by depression)		
	HAM-A 14-item	HAM-A psychic anxiety	HAM-A somatic anxiety	HAM-A 14-item	HAM-A psychic anxiety	HAM-A somatic anxiety
Age (sub/supramedian)	0.15	0.12	0.23	0.11	0.30	0.40
Sex (male/female)	0.10	0.09	0.12	0.01	0.23	0.03
Treatment status (in/outpatient)	0.05	0.12	0.06	–	–	–
<i>Diagnosis</i>						
With/without major depressive episode	0.002	0.02	0.12	–	–	–
With/without agoraphobia	0.03	0.11	0.34	–	–	–
With/without panic attacks	–	–	–	0.71	0.31	0.40
<i>Scale</i>						
HAM-D (sub/supramedian)	0.003	0.12	0.09	0.05	0.14	0.12
HAM-A (sub/supramedian)	0.001	0.08	0.13	0.007	0.31	0.06
HAM-A psychic (sub/supramedian)	0.01	0.11	0.09	0.004	0.16	0.16
HAM-A somatic (sub/supramedian)	0.06	0.08	0.03	0.003	0.16	0.42
Chance	0.34	0.39	0.63	0.08	0.32	0.60

TABLE 4

MEANS AND STANDARD ERRORS (IN BRACKETS) OF THE TOTAL SCORES OF THE HAM-A AND ITS SUBSCALES FOR DSM-III SUBGROUPS IN BOTH SAMPLES

All patients in this group had panic attacks. MDE = major depressive episode.

Sample and DSM-III-R subgroup	Scale			
	Number of patients	HAM-A global score	HAM-A psychic anxiety, global score	HAM-A somatic anxiety, global score
<i>Sample 1</i>				
Panic disorder without agoraphobia and without MDE	36	14.8 (1.1)	8.0 (0.8)	6.8 (0.6)
Panic disorder with agoraphobia without MDE	16	21.8 (1.6)	11.5 (0.9)	10.3 (0.8)
Panic disorder with MDE without agoraphobia	31	22.2 (1.4)	13.1 (0.8)	9.2 (0.8)
Panic disorder with agoraphobia and MDE	14	29.2 (1.4) **	15.3 (0.9) **	13.9 (0.9) **
<i>Sample 2</i>				
MDE without anxiety disorder	73	14.9 (0.8)	10.3 (0.6)	4.6 (0.5)
MDE with anxiety disorder	28	17.4 (1.8)	11.5 (1.2)	5.9 (0.9)

* Significant differences between subtypes by Kruskal–Wallis or *U*-test ($0.01 < P < 0.05$).** Significant differences between subtypes by Kruskal–Wallis or *U*-test ($P < 0.01$).

TABLE 5

MEANS AND STANDARD ERRORS (IN BRACKETS) OF THE HAM-A SCORES DURING TREATMENT IN THE TREATMENT TRIALS

Sample	Scale		
	HAM-A global score	HAM-A psychic anxiety, global score	HAM-A somatic anxiety, global score
<i>Sample 1</i>			
Pre treatment	25.4 (1.6)	14.0 (0.9)	11.3 (0.7)
Post treatment	17.0 (1.7) **	9.0 (0.9) **	8.0 (0.8) **
<i>Sample 2</i>			
Pre treatment	22.0 (1.1)	13.9 (0.8)	7.9 (0.7)
Post treatment	16.8 (1.6) **	9.5 (1.0) **	7.3 (0.7)

* Significant improvement by the Wilcoxon test ($0.01 < P < 0.05$).

** Significant improvement by the Wilcoxon test ($P < 0.01$).

In the sample of depressed patients (sample 2), the correlation between the anxiolytic effects measured by the HAM-A somatic anxiety score and the Covi score is rather low (0.33) (Table 6). The change measured with the somatic anxiety subscale was, however, significantly correlated with the number and intensity of side effects rated by the AMDP system (0.31) (Table 6).

TABLE 6

RELATIONSHIP (PEARSON CORRELATION) BETWEEN THE CHANGE SCORES OF THE HAM-A AND GLOBAL EXPERT ASSESSMENT OF THE ANXIOLYTIC AND ANTIDEPRESSANT EFFECT AND SIDE EFFECTS IN TWO TREATMENT TRIALS

The correlations are calculated for the residuals derived by regression analysis (observed values minus values expected on the basis of initial values).

Sample	Scale		
	HAM-A global score	HAM-A psychic anxiety, global score	HAM-A somatic anxiety, global score
<i>Sample 1</i>			
Global anxiety rating	0.73 **	0.76 **	0.65 **
<i>Sample 2</i>			
Global anxiety rating	0.59 **	0.79 **	0.33 **
Global depression rating	0.58 **	0.79 **	0.21 *
Side effects	0.10	-0.03	0.31 **

* Correlation coefficient significantly different from zero (t -test, $0.01 < P < 0.05$).

** Correlation coefficient significantly different from zero (t -test, $P < 0.01$).

Discussion

Several disadvantages of the HAM-A have been detected. First, the insufficient homogeneity and transferability: the global score of the HAM-A does not identify a stable latent variable. Similarly, Gjerris et al. (1983) found only a low degree of homogeneity. Therefore, more than one dimension is necessary for joining the item scores together. Second, the high degree of overlap between the anxiolytic effects measured by HAM-A change scores and the antidepressant effects: a lower degree of overlap is observed for the global change scores in depression (Raskin) and anxiety (Covi) scales. Therefore, more specific HAM-A items are required.

On the other hand, the HAM-A demonstrated:

- (1) Sufficient reliability except for a few items. The examination of reliability was carried out by correlating two ratings performed for each patient. According to the manual of the HAM-A (Hamilton, 1969), two raters should perform the rating and discuss their discrepancies afterwards in order to come to an agreement; this procedure will probably enhance the reliability of the HAM-A. We did not follow this rationale because usually one rater performs the assessment of the HAM-A; therefore, our setting is able to give a realistic estimate of the reliability of the HAM-A.

- (2) Satisfactory correlation coefficients with concurrent assessments of the cross-sectional severity of anxiety. Despite the low specificity for anxiolytic effects, the sensitivity to change during anxiolytic treatment is rather good; sufficient sensitivity to change has also been reported by Kellner et al. (1968).

However, these positive results do not outweigh the disadvantages.

The HAM-A was designed for patients with anxiety neurosis. However, it is also applied in depression. The HAM-A was unable to detect anxiety states in our sample of depressed patients. This lack of specificity does not justify using the HAM-A global score as an indicator of the severity of anxiety states in depressive disorders.

The HAM-A subscale of psychic anxiety was a better indicator of the severity of anxiety: the subscale was homogeneous and transferable in both samples studied. In spite of this, a lack of specificity in measuring anxiolytic effects of this subscale can be observed in the sample of depressed patients; therefore, this subscale can hardly be used as an indicator of the severity of anxiety states in depressive disorders.

A reason for the low specificity of the HAM-A, demonstrated by a high degree of overlap with features of depression, is due to some degree to the inclusion of an item measuring the severity of depressive states. But in spite of the high degree of overlap between anxiety and depression in epidemiological and clinical samples, a clear distinction between anxiety and depression is needed in clinical psychopharmacology; for antidepressant effects must be distinguished from anxiolytic effects (Lader, 1985). Otherwise, antidepressant drugs may be misidentified as anxiolytics and vice versa. This problem cannot be solved by restricting the applicability of the HAM-A to anxiety disorders; for anxiolytic drugs are clinically effective in the treatment of anxiety in anxiety disorders as well as in depressive disorders. Therefore, the degree of anxiety must also be assessed when testing the efficacy of anxiolytics in depressive disorders. Similarly, the features of depression must be separated from the features of anxiety in anxiety disorders because of their different impact on treatment. A scale intended to be an indicator of the efficacy of anxiolytics has to find a solution

for all these problems. This is difficult because no specific and generally accepted external indicators of anxiety (e.g., psychophysiological variables) are available. Empirical psychopathological studies dedicated to the discrimination between anxiety and depression are the only basis for finding the most specific psychopathological indicators of anxiety states. But the HAM-A is not in accordance with the current research in this field. For example, panic attacks, which are one of the most specific indicators of anxiety states (Mountjoy and Roth, 1982), are not included in this scale.

Another problem arises with the subscale of somatic anxiety. In spite of being homogeneous and transferable, the change scores of this subscale represent improvement of somatic anxiety as well as the appearance of new side effects of the treatment. This disadvantage can be explained by also counting somatic symptoms in the HAM-A when anxiety is not associated with the somatic complaints. We have only been able to consider the impact of side effects on the somatic subscale in patients defined by the presence of a major depressive episode. However, this problem may be relevant in anxiety disorders, too.

A further reason for the limitations in validity is the conceptualization of anxiety in the HAM-A: no advice is given on how to balance severity and duration, e.g., how to balance the severity of symptoms limited to short time intervals (panic attacks) and the severity of enduring symptoms. Matching these points would increase the reliability and the validity of the HAM-A.

The insufficient specificity of the HAM-A for the assessment of change in the severity of anxiety will limit its applicability in treatment evaluation studies. Revisions of the HAM-A or new scales are needed, especially for treatment trials with anxiolytics. Crucial requirements for scales used for this purpose are: sufficient homogeneity and transferability, sufficient specificity and sensitivity for anxiety states and the ability to detect change in the severity of anxiety.

Several authors have proposed promising procedures for establishing new anxiety scales characterized by a higher specificity than the HAM-A. Snaith et al. (1982) proposed a scale comprising the core symptoms of anxiety (items from the HAM-A subscale of psychic anxiety and other

PSE-related items) and excluding items referring to depressive states. Bech et al. (1986) proposed using the HAM-A in a twofold manner: the assessment of panic anxiety is separated from the assessment of generalized anxiety; both assessments are performed separately by the original version of the HAM-A. Sheehan et al. (1983) proposed a series of new scales: one scale for panic attacks, one scale for phobia.

Unfortunately, none of these procedures has been tested for its reliability and validity. We also have not been able to include those recently proposed procedures in our studies. The limitations of the validity of the original HAM-A version documented by our data underline the necessity to include these alternative instruments in forthcoming treatment trials in order to collect data able to validate these scales.

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